La Scuola d'Italia / Dr. Huson / IB Math
27 April 2026

First & Last name:

**7.3 Classwork: Logarithm and Graphing Challenge (no calculator)**

1. **Change of base**

**(a)** Solve for $x$:

$$\log_2 x \;+\; \log_4 x \;+\; \log_8 x \;=\; 11.$$

**(b)** Show that, for $a, b > 0$ with $a, b \neq 1$,

$$\log_a b \cdot \log_b a \;=\; 1.$$

**(c)** Given that $\log_a b = 4$, find the exact value of $\log_{a^2} b^3$.

2. **Quadratic in log**

   *Hint: substitute* $u = \log_3 x$ *to obtain a quadratic in* $u$.

   Solve for $x$:
   $$(\log_3 x)^2 \;-\; 5\log_3 x \;+\; 6 \;=\; 0.$$

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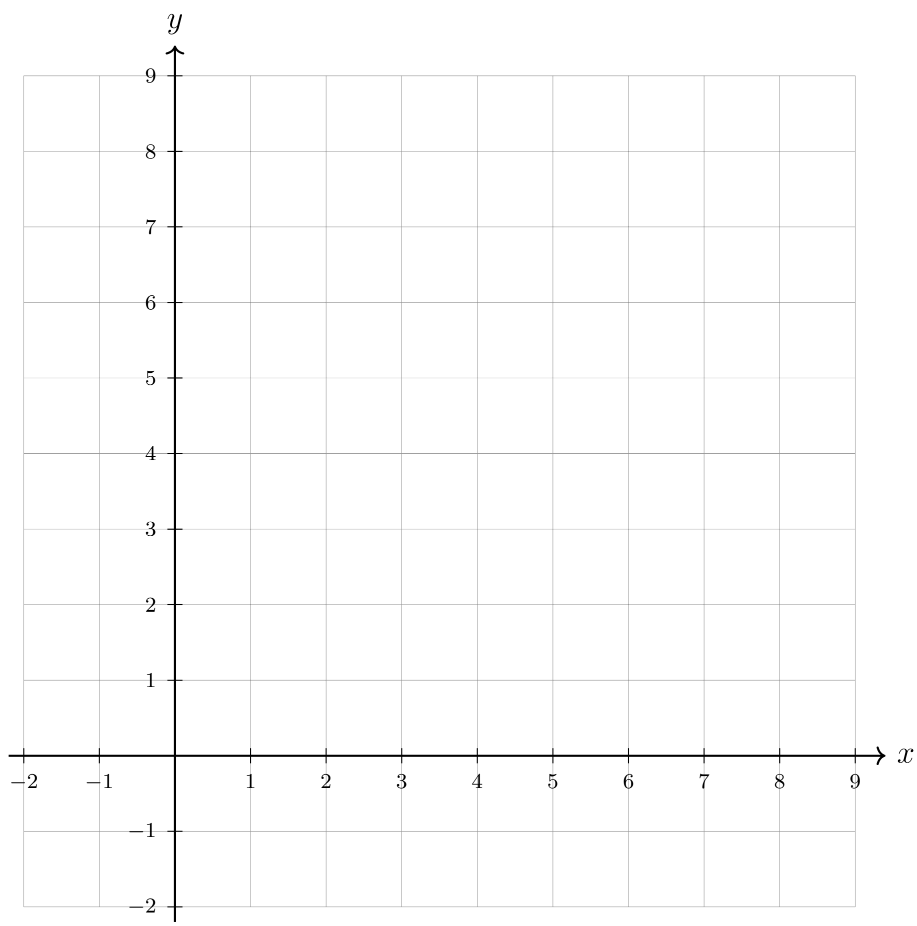
3. **An exponential and its inverse**

Let $f(x) = 3^{x-1} + 2$.

**(a)** Find $f^{-1}(x)$.

**(b)** On the axes below, sketch $y = f(x)$, $y = f^{-1}(x)$, and $y = x$. Label each curve and the asymptote of each.



4. **Solve for the exponent**

When $x$ is in the exponent, take the logarithm of both sides. Give exact answers.

**(a)** Solve $2^x = 20$.

**(b)** Solve $5^{x-1} = 12$.

**(c)** A bacteria culture is modelled by $P(t) = 50 \cdot 2^t$, where $t$ is the time in hours. Find the exact value of $t$ at which $P = 1500$.

**(d)** Solve $5^{2x} = 3^{x+1}$.